Representations of data

Experimental

i.e. Dealing with collected data.

Chp1: Data

Methods of sampling, types of data, and populations vs samples.

Chp2: Measures of Location/Spread

Statistics used to summarise data, including mean, standard deviation, quartiles, percentiles. Use of linear interpolation for estimating medians/quartiles.

Chp3: Representation of Data

Producing and interpreting visual representations of data, including box plots and histograms.

Chp4: Correlation

Measuring how related two variables are, and using linear regression to predict values.

Theoretical

Deal with probabilities and modelling to make inferences about what we 'expect' to see or make predictions, often using this to reason about/contrast with experimentally collected data.

Chp5: Probability

Venn Diagrams, mutually exclusive + independent events, tree diagrams.

Chp6: Statistical Distributions

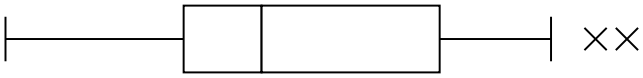
Common distributions used to easily find probabilities under certain modelling conditions, e.g. binomial distribution.

Chp7: Hypothesis Testing

Determining how likely observed data would have happened 'by chance', and making subsequent deductions.

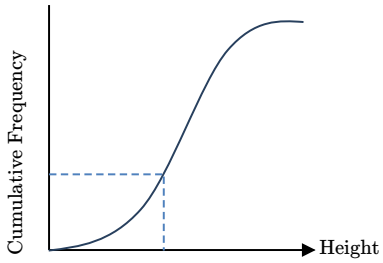
We've seen so far how data is collected and calculations can be made. We now concentrate on how the processed data can be *displayed*.

BOX PLOTS AND OUTLIERS

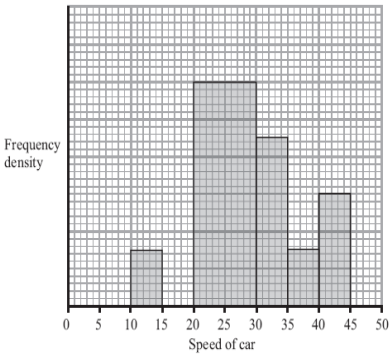


NEW since GCSE! Outliers.

CUMULATIVE FREQ DIAGRAMS



HISTOGRAMS



NEW since GCSE!
Area is not necessarily equal to frequency.
Forming a frequency polygon by joining midpoints.

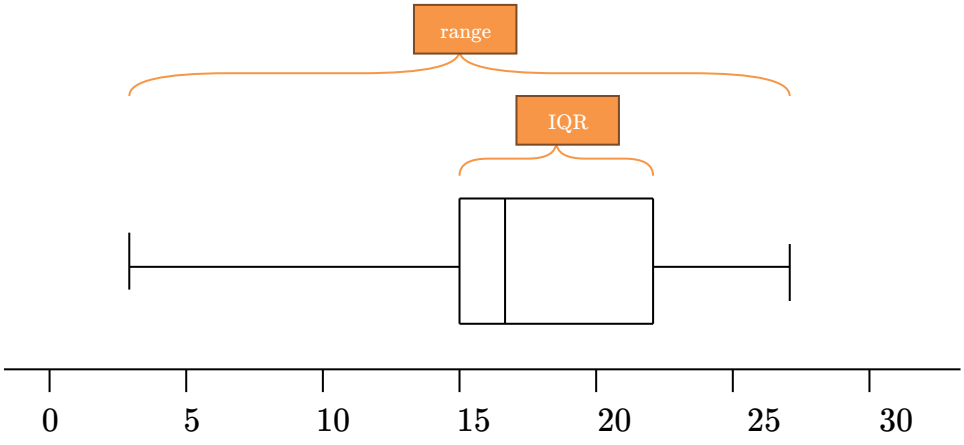
Changes since the old 'S1' syllabus:

- Stem and leaf diagrams have been cut. 'Skew' has been cut.
- Cumulative frequency diagrams have been added.
- Turning histogram into frequency polygon.

Box Plots allow us to visually represent the distribution of the data.

Minimum	Lower Quartile	Median	Upper Quartile	Maximum
3	15	17	22	27

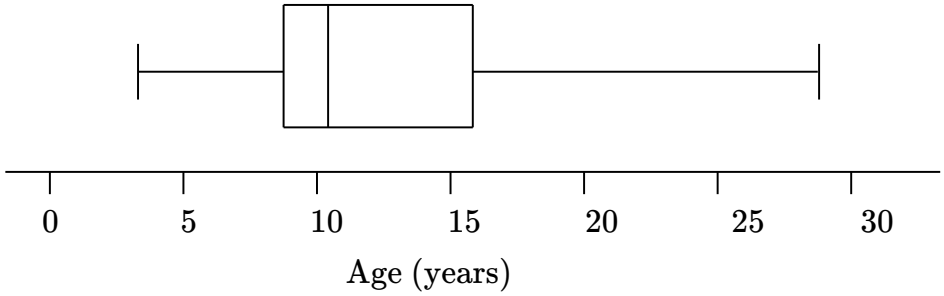
Sketch Sketch Sketch Sketch Sketch



How is the **IQR** represented in this diagram? Sketch

How is the **range** represented in this diagram? Sketch

Interpreting a Box Plot 5



True or false: (click your answer)

"The right box represents more people than the left box."

False True

Each box represents 25% of people, i.e. the same number of people!

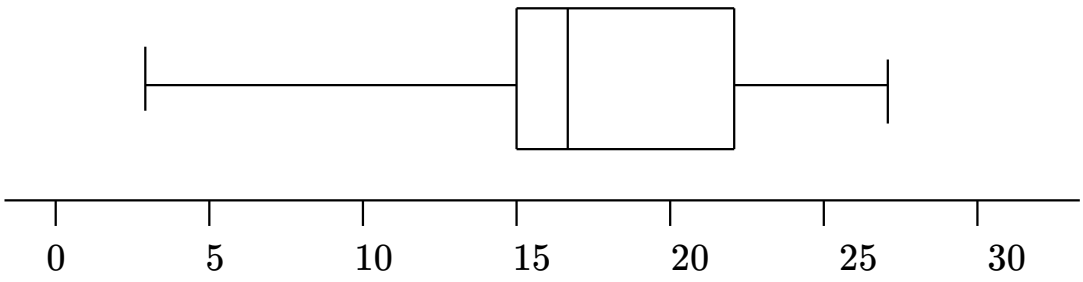
"The ages are more spread out above the median."

False True

The wider the box or whisker, the more spread out the values are within that 25% of the data. We'd say that the data has "**positive skew**", but you are not required to know this term.

Outliers 6

An outlier is: ?



The diameters of 11 different Roman coins are measured in centimetres:

2.2 2.5 2.7 2.7 2.8 3.0 3.1 3.1 3.2 4.0 4.7

Determine the quartiles and hence any outliers.

?

[Textbook] The lengths, in cm, of 12 giant African land snails are given below:

17 18 18 19 20 20 20 20 21 23 24 32

a) Calculate the mean and standard deviation, given that $\Sigma x = 252$ and $\Sigma x^2 = 5468$.

b) An outlier is an observation which lies ± 2 standard deviations from the mean. Identify any outliers for this data.

? a

? b

Context: Recall that the standard deviation is, roughly speaking, the average distance of each value from the mean. So the outlier definition is saying we're at least twice this average distance, which seems like a sensible definition.

In Year 2, you will encounter the **normal distribution**, which can be used to model data which is **clustered about some mean and tails off symmetrical in either direction**. If this data was approximately normally distributed, then there is a 5% chance a random observation would fall outside 2 standard deviations within the mean. You will learn then how to make such probability calculations.

The ages of 15 Lib Dem MPs are given:

11 18 20 27 30 31 32 32 35 36 37 58 63 78 105

a) If an outlier is considered to be 1.5 interquartile ranges below the lower quartile or above the upper quartile, determine any outliers.

b) If instead an outlier is considered to be outside 2 standard deviations within the mean, determine any outliers. Note that $\Sigma x = 613$ and $\Sigma x^2 = 33815$

? a

Therefore the 105 year old is an outlier.

? b

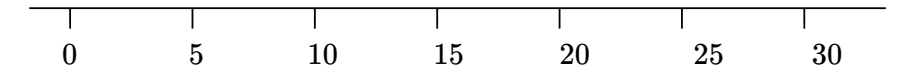
Smallest values	Largest values	Lower Quartile	Median	Upper Quartile
0, 3	21, 27	8	10	14

Draw a box plot to represent the above data.

?

Exam Tip: You MUST show your outlier boundary calculations.

?



[Jan 2011 Q3] Over a long period of time a small company recorded the amount it received in sales per month. The results are summarised below.

	Amount received in sales (£1000s)
Two lowest values	3, 4
Lower quartile	7
Median	12
Upper quartile	14
Two highest values	20, 25

An outlier is an observation that falls
either 1.5 × interquartile range above the upper quartile
or 1.5 × interquartile range below the lower quartile.

(*a*) On the graph paper below, draw a box plot to represent these data, indicating clearly any outliers. **(5)**

(a) a ?

(c) The company claims that for 75% of the months, the amount received per month is greater than £10 000. Comment on this claim, giving a reason for your answer. **(2)**

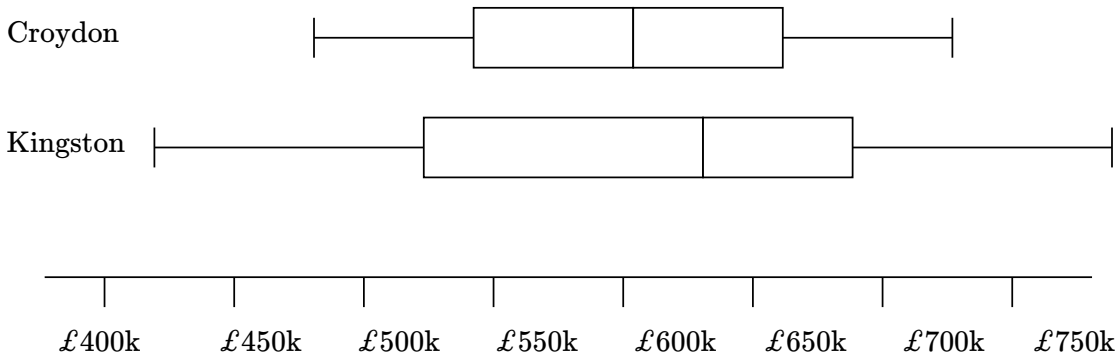
M1
A1

M1

A1ft (c) c ?

B1

Box Plot comparing house prices of Croydon and Kingston-upon-Thames:



"Compare the prices of houses in Croydon with those in Kingston". (2 marks)

For 1 mark, one of:

?

For 1 mark:

?

4 The ages of nine people at a children's birthday party are recorded. $\Sigma x = 92$ and $\Sigma x^2 = 1428$.

a Calculate the mean and standard deviation of the ages. **(3 marks)**

An outlier is an observation which lies ±2 standard deviations from the mean.

One of the ages is recorded as 30.

b State, with a reason, whether this is an outlier. **(2 marks)**

c Suggest a reason why this age could be a legitimate data value. **(1 mark)**

d Given that all nine people were children, clean the data and recalculate the mean and standard deviation. **(3 marks)**

a?

b?

c?

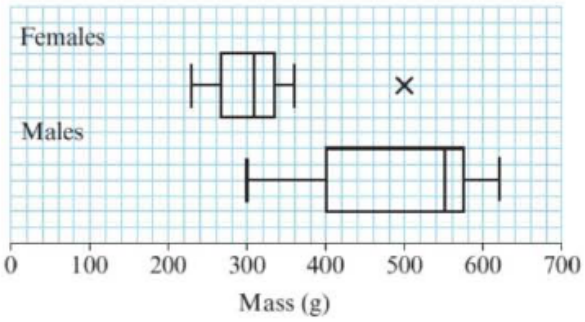
d?

3 The masses of male and female turtles are given in grams. Their masses are summarised in the box plots.

a Compare and contrast the masses of the male and female turtles.

b A turtle was found to have a mass of 330 grams. State whether it is likely to be a male or a female. Give a reason for your answer.

c Write down the size of the largest female turtle.



a?

b?

c?

4 Data for the maximum daily gust (in knots) in Camborne in September 1987 is taken from the large data set:

13	17	19	20	21
21	22	23	24	25
25	25	26	26	26
27	29	30	30	30
33	35	38	46	78

© Crown Copyright Met Office

a Calculate Q_1, Q_2 and Q_3. **(3 marks)**

An outlier is defined as a value which lies either 1.5 × the interquartile range above the upper quartile or 1.5 × the interquartile range below the lower quartile.

b Show that 46 and 78 are outliers. **(1 mark)**

c Draw a box plot for this data. **(3 marks)**

a?

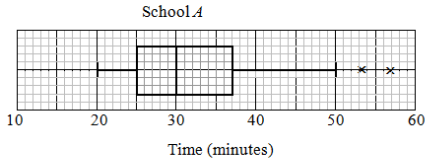
b?

c?

5. [May 2006 Q1] (*a*) Describe the main features and uses of a box plot. **(3)**

Children from schools *A* and *B* took part in a fun run for charity. The times, to the nearest minute, taken by the children from school *A* are summarised in Figure 1.

Figure 1



(*b*) (i) Write down the time by which 75% of the children in school *A* had completed the run.
(ii) State the name given to this value. **(2)**
(*c*) Explain what you understand by the two crosses (×) on Figure 1. **(2)**

For school *B* the least time taken by any of the children was 25 minutes and the longest time was 55 minutes. The three quartiles were 30, 37 and 50 respectively.
(*d*) On graph paper, draw a box plot to represent the data from school *B*. **(4)**
(*e*) Compare and contrast these two box plots. **(4)**

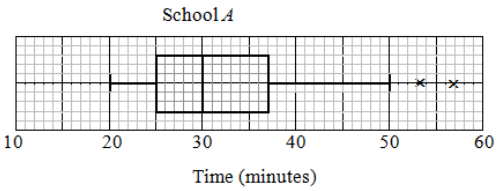
(Solutions to (d) and (e) on next slide)

1(a)	?	B1 B1 B1 (3)
(b)(i)	?	B1
(ii)		B1 (2)
(c)	?	B1 B1 (2)

5. [May 2006 Q1] (*a*) Describe the main features and uses of a box plot. **(3)**

Children from schools *A* and *B* took part in a fun run for charity. The times, to the nearest minute, taken by the children from school *A* are summarised in Figure 1.

Figure 1



(*b*) (i) Write down the time by which 75% of the children in school *A* had completed the run.
(ii) State the name given to this value. **(2)**
(*c*) Explain what you understand by the two crosses (×) on Figure 1. **(2)**

For school *B* the least time taken by any of the children was 25 minutes and the longest time was 55 minutes. The three quartiles were 30, 37 and 50 respectively.
(*d*) On graph paper, draw a box plot to represent the data from school *B*. **(4)**
(*e*) Compare and contrast these two box plots. **(4)**

(d)	?	
(e)	?	B1 B1 B1 B1

6. [June 2005 Q4] Aeroplanes fly from City *A* to City *B*. Over a long period of time the number of minutes delay in take-off from City *A* was recorded. The minimum delay was 5 minutes and the maximum delay was 63 minutes. A quarter of all delays were at most 12 minutes, half were at most 17 minutes and 75% were at most 28 minutes. Only one of the delays was longer than 45 minutes.

An outlier is an observation that falls either 1.5 × (interquartile range) above the upper quartile or 1.5 × (interquartile range) below the lower quartile.

(*a*) On graph paper, draw a box plot to represent these data. **(7)**
(*b*) Comment on the distribution of delays. Justify your answer. **(2)**
(*c*) Suggest how the distribution might be interpreted by a passenger who frequently flies from City *A* to City *B*. **(1)**

a?

b?

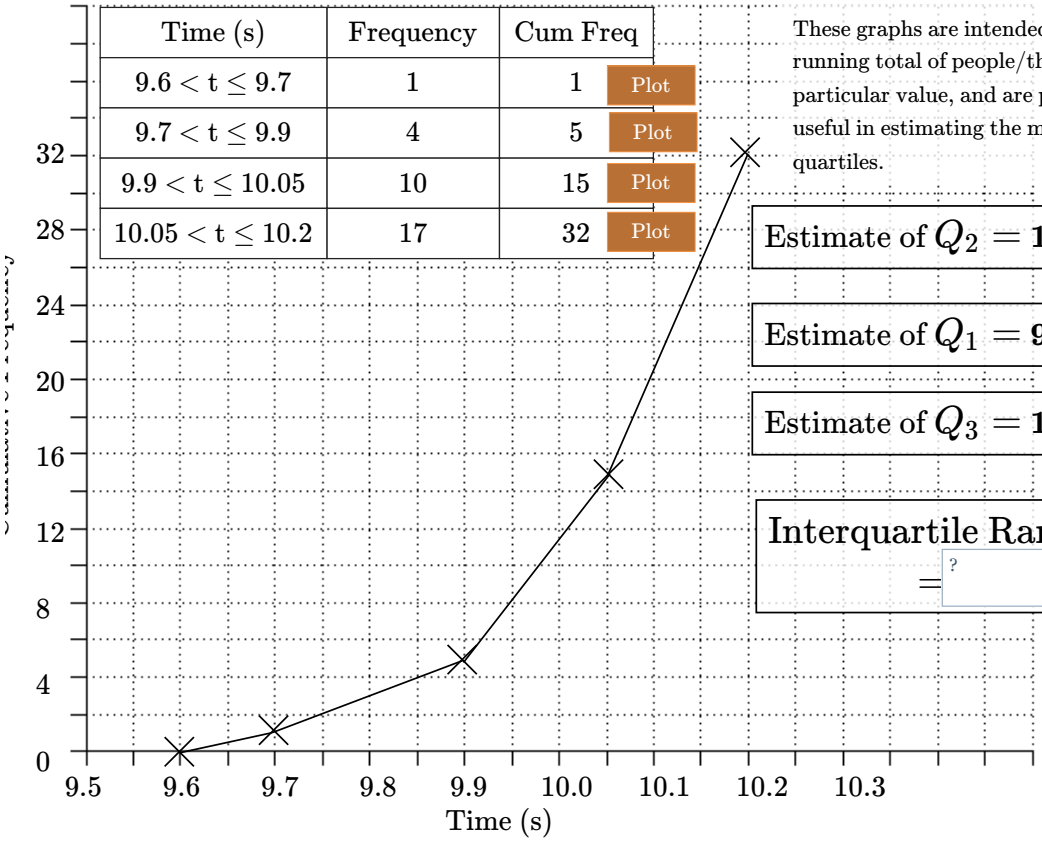
c?

B1; B1 (2)

B1 (1)

These graphs are intended to show the running total of people/things up to a particular value, and are particularly useful in estimating the median and quartiles. 18

Time (s)	Frequency	Cum Freq	
$9.6 < t \leq 9.7$	1	1	Plot
$9.7 < t \leq 9.9$	4	5	Plot
$9.9 < t \leq 10.05$	10	15	Plot
$10.05 < t \leq 10.2$	17	32	Plot



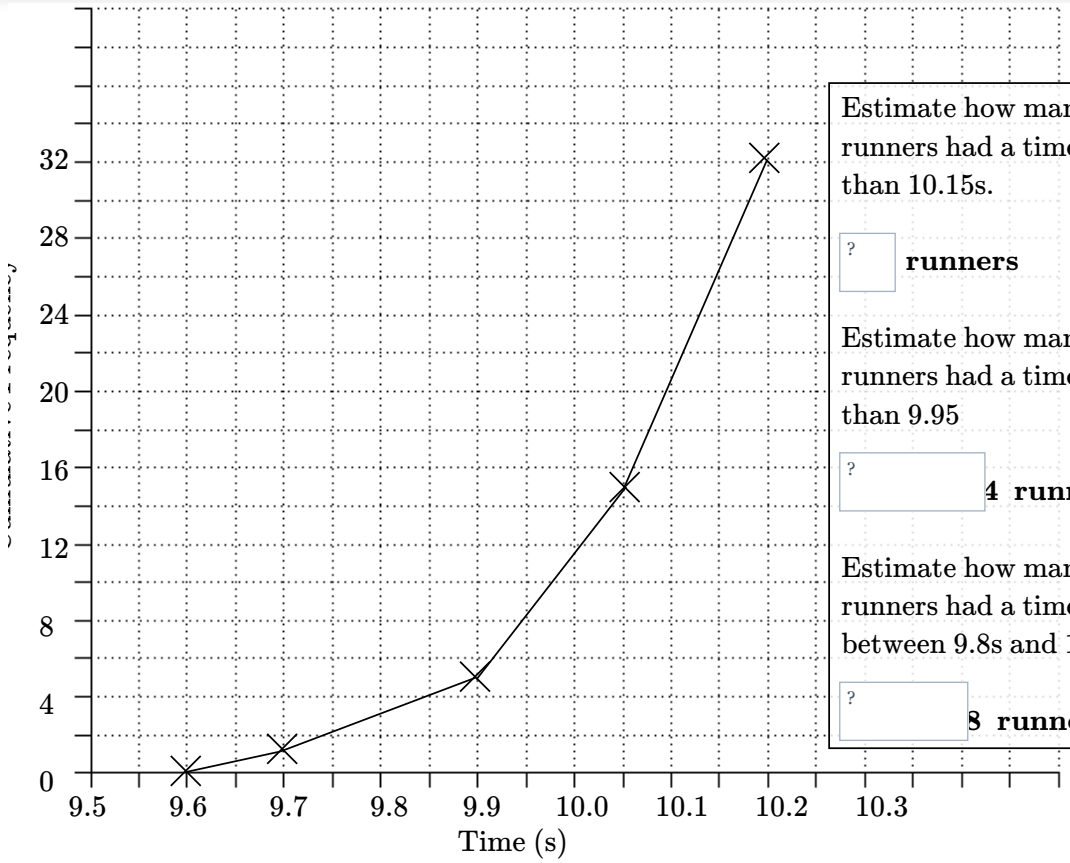
These graphs are intended to show the running total of people/things **up to** a particular value, and are particularly useful in estimating the median and quartiles.

Estimate of $Q_2 = 10$?

Estimate of $Q_1 = 9.$?

Estimate of $Q_3 = 10$?

Interquartile Range = ?



Estimate how many runners had a time less than 10.15s.

? **runners**

Estimate how many runners had a time more than 9.95

? **4 runners**

Estimate how many runners had a time between 9.8s and 10s

? **8 runners**

4 A vet measures the masses, in kg, of male and female domestic shorthair cats. Her results are given in the table.

Mass, w (kg)	Frequency (male)	Frequency (female)
$2.5 \leqslant w < 3.0$	1	5
$3.0 \leqslant w < 3.5$	12	17
$3.5 \leqslant w < 4.0$	20	32
$4.0 \leqslant w < 4.5$	27	12
$4.5 \leqslant w < 5.0$	7	4
$5.0 \leqslant w < 5.5$	3	0

a Draw cumulative frequency diagrams for both sets of data on the same axes.

b Which gender has the greater spread of masses?

A female domestic shorthair cat is considered underweight if its mass is below 3.2 kg.

A male domestic shorthair cat is considered underweight if its mass is below 3.8 kg.

c Which gender has fewer underweight cats?

a?

b?

c?

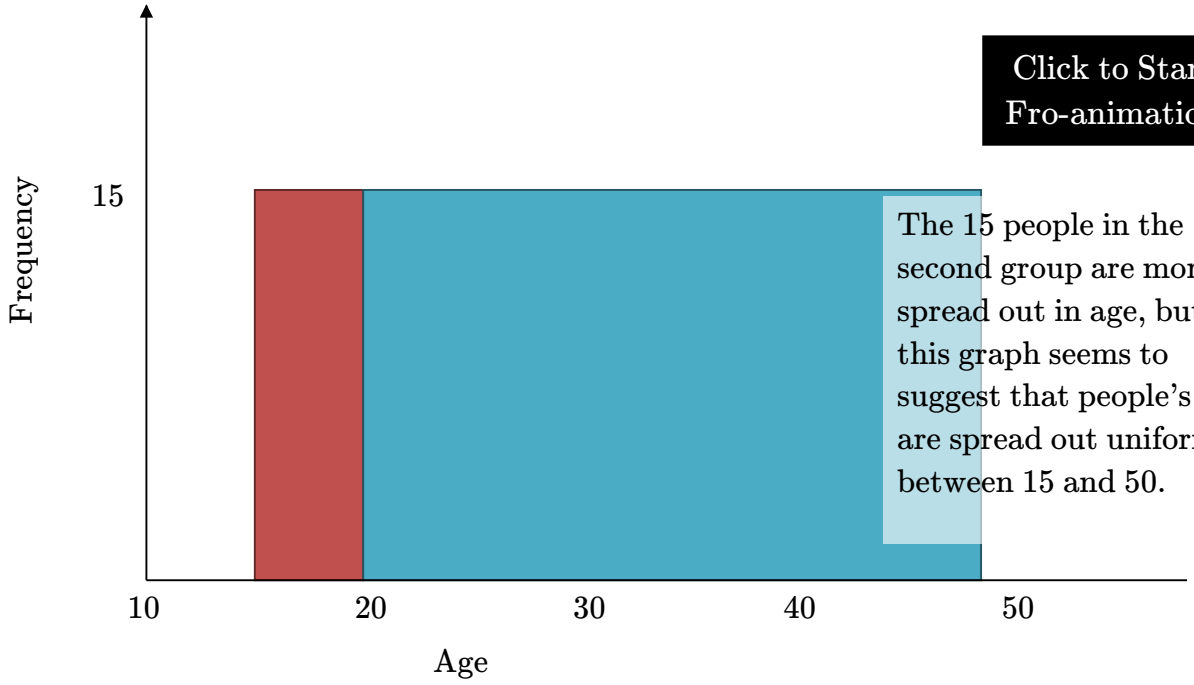
Pablo is hosting a party. He counts how many people are between 15 and 20, and 20 and 50.

Age (years)	Frequency
$15 \leq a < 20$	15
$20 \leq a < 50$	15

Pablo is hosting a party. He counts how many people are between 15 and 20, and 20 and 50.

Why is below graph somewhat unhelpful. How could we fix it?

Click to Start Fro-animation



The 15 people in the second group are more spread out in age, but this graph seems to suggest that people's ages are spread out uniformly between 15 and 50.

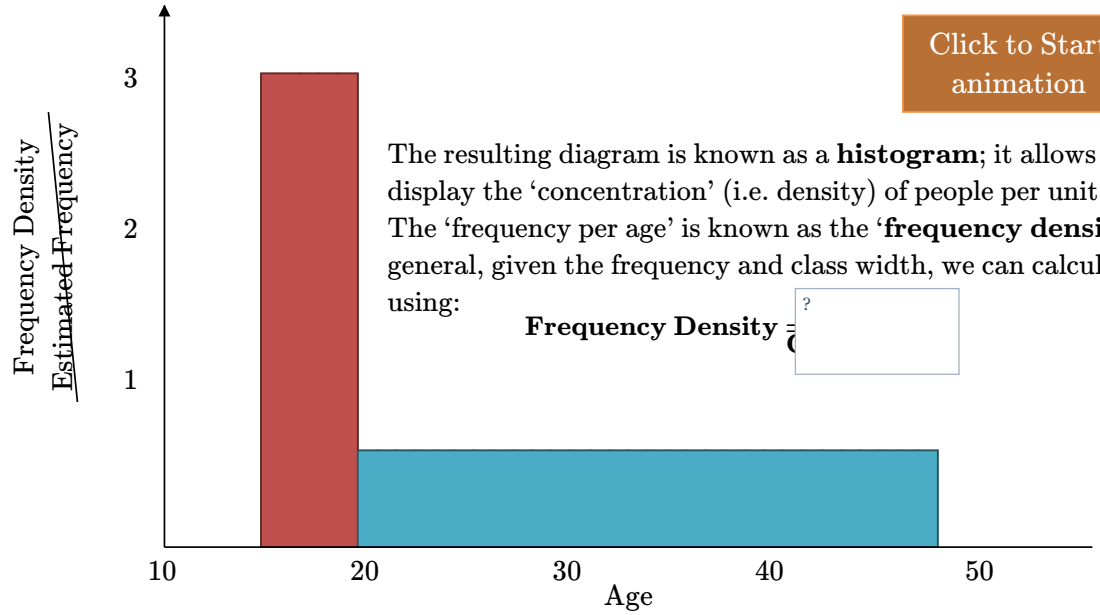
Estimating frequency within an age group

Age (years)	Frequency
$15 \leq a < 20$	15
$20 \leq a < 50$	15

Let's presume that within each age group, the ages are evenly spread.

Then there would **3** people of each age in the 15-20 group, and **0.5** people of each age in the 20-50 group.

Click to Start animation



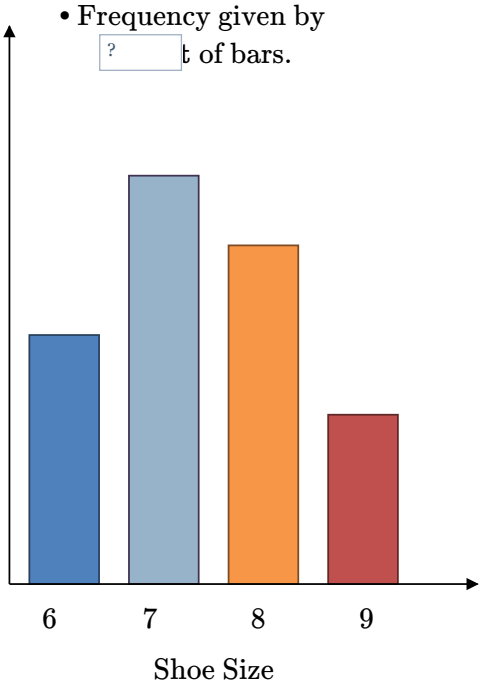
The resulting diagram is known as a **histogram**; it allows us to display the 'concentration' (i.e. density) of people per unit value. The 'frequency per age' is known as the '**frequency density**'. In general, given the frequency and class width, we can calculate it using:

Frequency Density =

Bar Charts vs Histograms

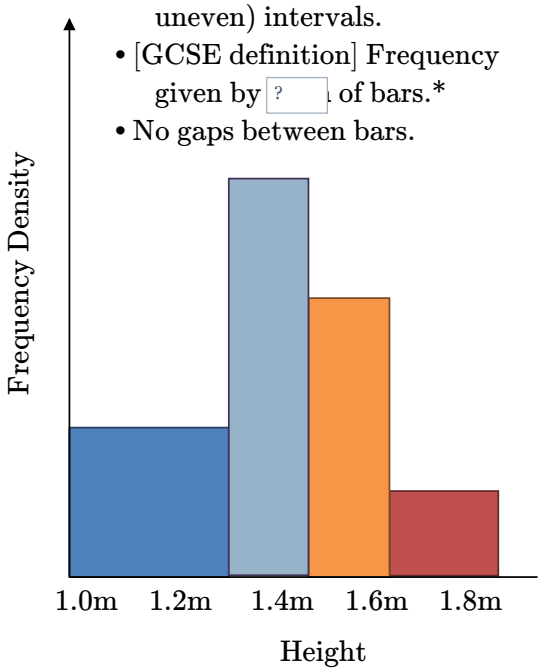
Bar Charts

- For d[?] data.
- Frequency given by [?] t of bars.



Histograms

- **For c[?] data.**
- Data divided into (potentially uneven) intervals.
- [GCSE definition] Frequency given by [?] of bars.*
- No gaps between bars.



* Not necessarily true. We'll correct this in a sec.

Weight (w kg)

Q1

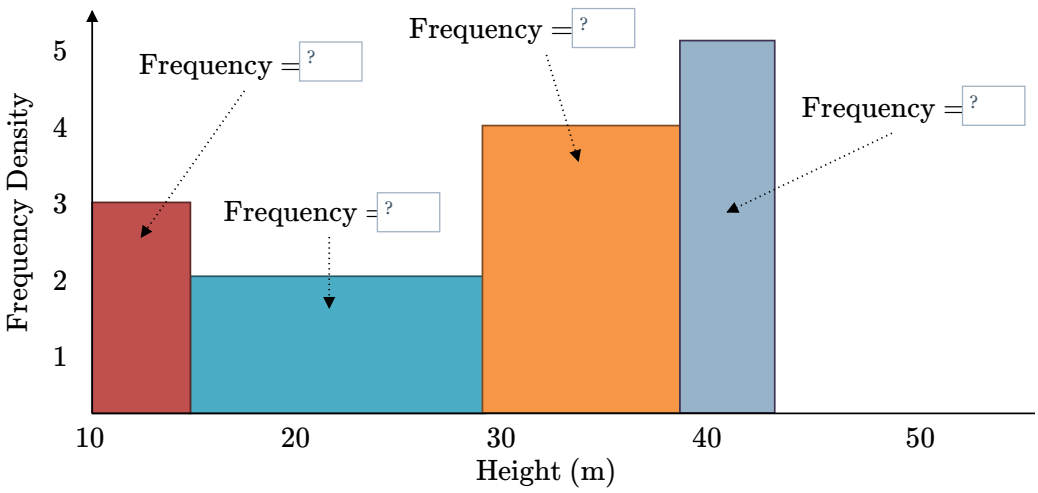
Weight (w kg)	Frequency	Frequency Density
$0 < w \leq 10$	40	[?]
$10 < w \leq 15$	6	[?]
$15 < w \leq 35$	[?]	2.6
$35 < w \leq 45$	[?]	1

Still using the **incorrect** GCSE formula:

Freq

F.D. Width

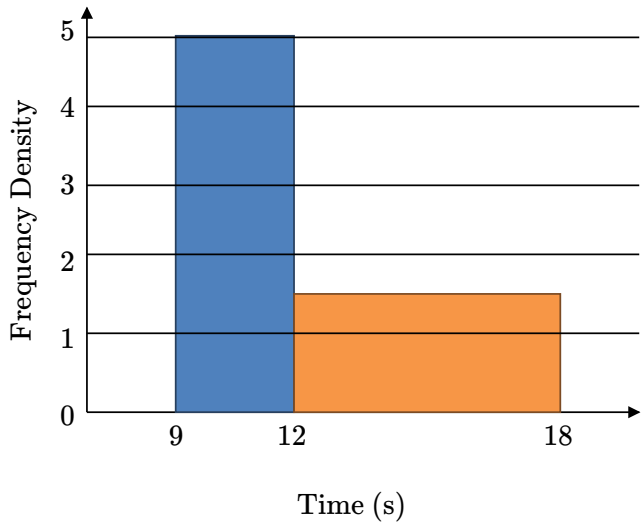
Q2



Unlike at GCSE, the area of a bar is not necessarily equal to the frequency; there are just **proportional**.

✎ Identify the scaling $area \times kfrequency$ using a known area with known frequency (which may be total area/frequency or just one bar)

There were 60 runners in a 100m race. The following histogram represents their times. Determine the number of runners with times above 14s.



Total frequency is known; therefore find total area and hence the 'scaling'.

?

Then use this scaling along with the desired area.

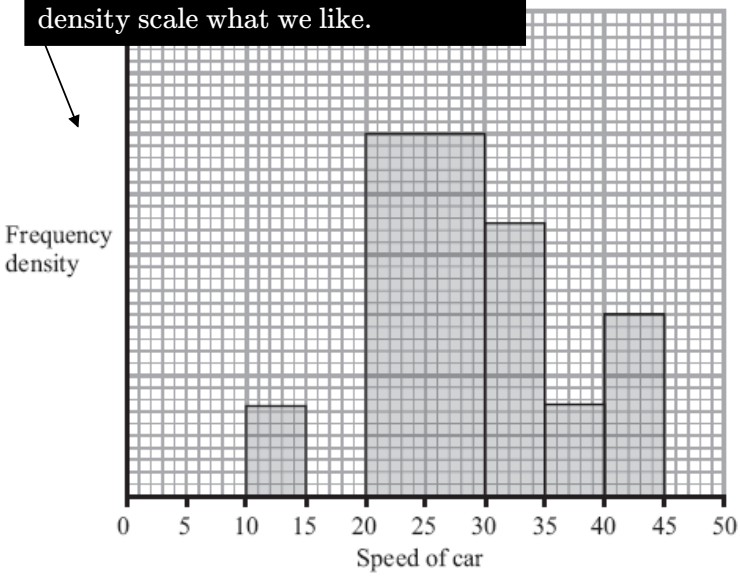
?

Edexcel S1 May 2012 Q5

A policeman records the speed of the traffic on a busy road with a 30 mph speed limit. He records the speeds of a sample of 450 cars. The histogram in Figure 2 represents the results.

(a) Calculate the number of cars that were exceeding the speed limit by at least 5 mph in the sample. *(4 marks)*

Tip: We can make the frequency density scale what we like.



M1 A1: Determine what one small square or one large square is worth.

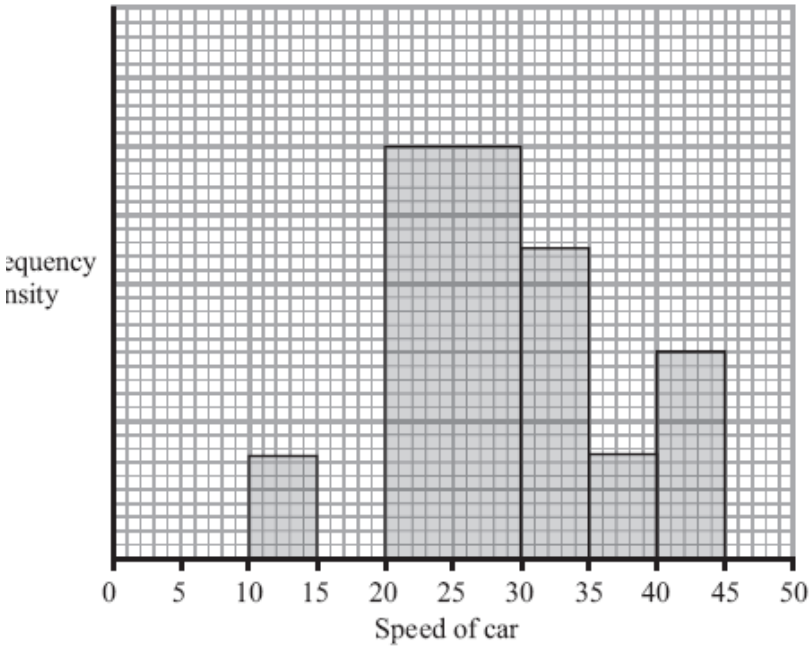
(i.e. work out $area \rightarrow freq$ scaling)

?

M1 A1: Use this to find number of cars travelling >35mph.

?

(b) Estimate the value of the mean speed of the cars in the sample. ***(3 marks)***



M1 M1: Use histogram to construct sum of speeds.

?

A1 Correct value

?

Tip: Whenever you are asked to calculate mean, median or quartiles from a histogram, form a grouped frequency table. Use your scaling factor to work out the frequency of each bar.

(*c*) Estimate, to 1 decimal place, the value of the median speed of the cars in the sample. **(2)**

(*d*) Comment on the shape of the distribution. Give a reason for your answer. **(2)**

(*e*) State, with a reason, whether the estimate of the mean or the median is a better representation of the average speed of the traffic on the road. **(2)**

(crossed out questions would not appear in new syllabus)

(c) ?

M1
A1 (2)

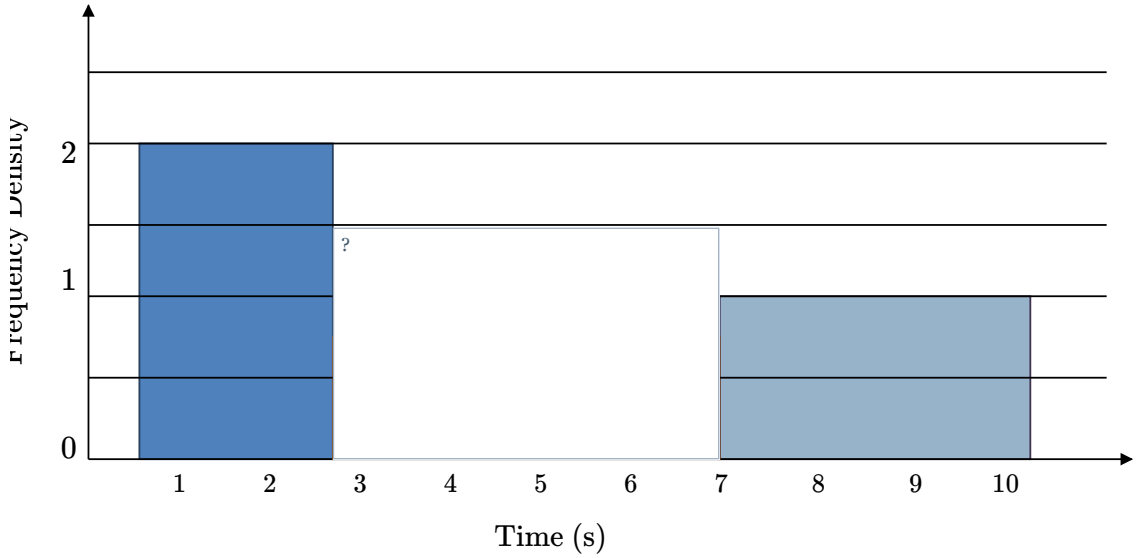
(d) ?

B1ft
dB1ft (2)

(e) ?

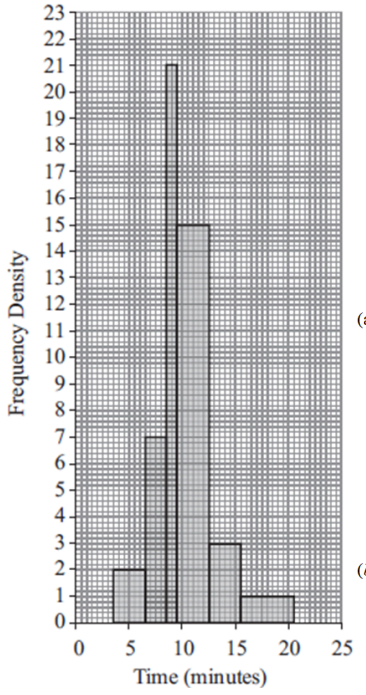
B1
dB1 (2)

Weight (to nearest kg)	Frequency	F.D.
1-2	4	?
3-6	3	?
7-9	?	?



Jan 2012 Q1

The histogram in Figure 1 shows the time, to the nearest minute, that a random sample of 100 motorists were delayed by roadworks on a stretch of motorway.



Tip: Be careful that you use the correct class widths!

(a) Complete the table.

Delay (minutes)	Number of motorists
4 – 6	6
7 – 8	?
9	21
10 – 12	45
13 – 15	9
16 – 20	?

(2)

(*b*) Estimate the number of motorists who were delayed between 8.5 and 13.5 minutes by the roadworks.

(2)

?

An exam favourite is to ask what width and height we'd draw a bar in a drawn histogram.

Q: The frequency table shows some running times. On a histogram the bar for 0-4 seconds is drawn with width 6cm and height 8cm. Find the width and height of the bar for 4-6 seconds.

Time (seconds)	Frequency
$0 \leq t < 4$	8
$4 \leq t < 6$	9

Tip Strategy?

Solution?

[May 2009 Q3] The variable x was measured to the nearest whole number. Forty observations are given in the table below.

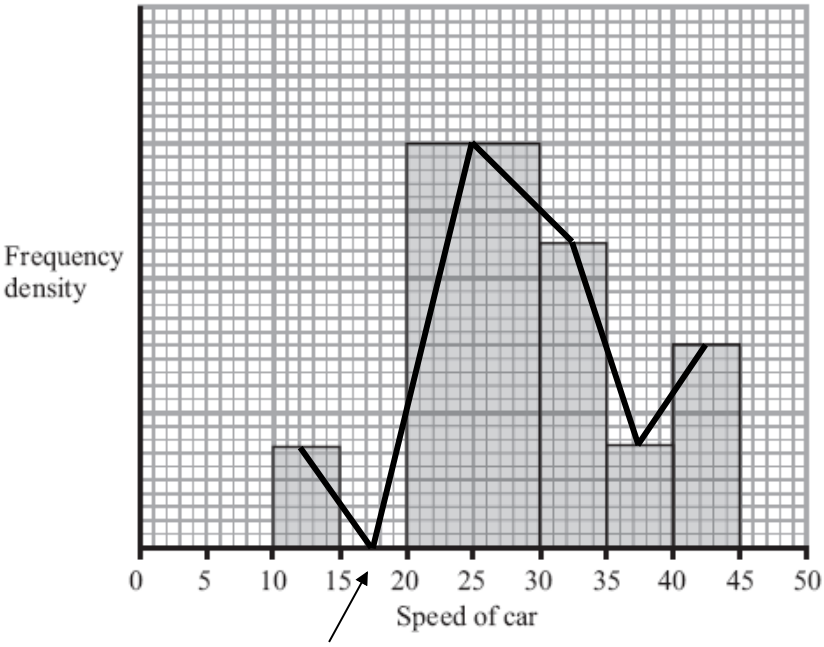
x	10 – 15	16 – 18	19 –
Frequency	15	9	16

A histogram was drawn and the bar representing the 10 – 15 class has a width of 2 cm and a height of 5 cm. For the 16 – 18 class find

(*a*) the width, **(1)**
(*b*) the height **(2)**
of the bar representing this class.

(a)	?	B1
(b)	?	M1 M1 A1

Recall that a frequency polygon can be drawn by using the midpoint of each interval. This corresponds to the midpoint of the top of each bar in a histogram.



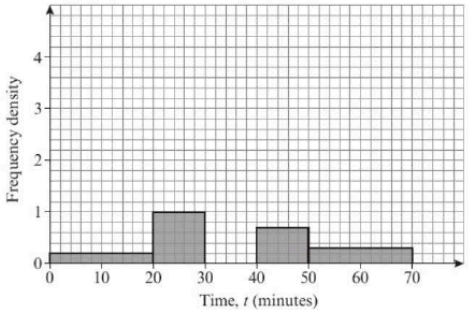
Click to Sketch

Note that the frequency in this interval is 0. That needs to be reflected in the frequency polygon.

5 The partially completed histogram shows the time, in minutes, that passengers were delayed at an airport.

a i Copy and complete the table.

Time, t (min)	Frequency
$0 \leqslant t < 20$	4
$20 \leqslant t < 30$	
$30 \leqslant t < 35$	15
$35 \leqslant t < 40$	25
$40 \leqslant t < 50$	
$50 \leqslant t < 70$	



ii Copy and complete the histogram. **(4 marks)**

b Estimate the number of passengers that were delayed for between 25 and 38 minutes. **(2 marks)**

a?

b?

6 The variable y was measured to the nearest whole number. 60 observations were taken and are recorded in the table below.

y	10–12	13–14	15–17	18–25
Frequency	6	24	18	12

a Write down the class boundaries for the 13–14 class. **(1 mark)**

A histogram was drawn and the bar representing the 13–14 class had a width of 4 cm and a height of 6 cm.

For the bar representing the 15–17 class, find:

b i the width **(1 mark)**

ii the height. **(2 marks)**

a?

b(i)?

b(ii)?

7 From the large data set, the daily mean temperature for Leeming during May 2015 is summarised in the table.

Daily mean temperature, t (°C)	Frequency
$4 \leqslant t < 8$	4
$8 \leqslant t < 10$	8
$10 \leqslant t < 11$	6
$11 \leqslant t < 12$	7
$12 \leqslant t < 15$	5
$15 \leqslant t < 16$	1

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A histogram was drawn. The $8 \leqslant t < 10$ group was represented by a bar of width 1 cm and a height of 8 cm.

a Find the width and height of the bar representing the $11 \leqslant t < 12$ group. **(2 marks)**

b Use your calculator to estimate the mean and standard deviation of temperatures in Leeming in May 2015. **(3 marks)**

c Use linear interpolation to find an estimate for the lower quartile of temperatures. **(2 marks)**

d Estimate the number of days in May 2015 on which the temperature was higher than the mean plus one standard deviation. **(2 marks)**

a?

b?

c?

d?

4 A sample of the daily maximum relative humidity is taken from the large data set for Leuchars and for Camborne during 2015. The data is given in the table.

Leuchars	100	98	100	100	100	100	100	100	94	100	91	100	100	89	100
Camborne	92	95	99	96	100	100	90	98	81	99	100	99	91	98	100

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a Find the median and quartiles for both samples. **(4 marks)**

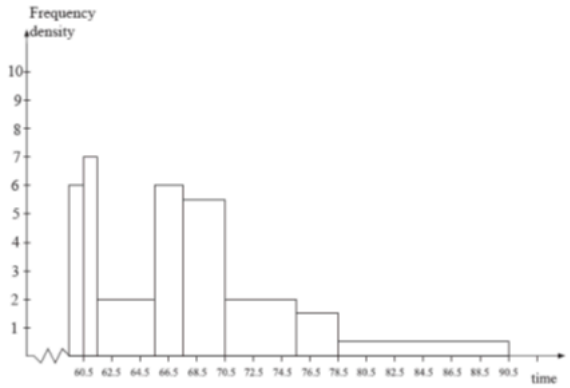
b Compare the two samples. **(2 marks)**

a?

b?

Q1

[Jan 2008 Q3] The histogram in Figure 1 shows the time taken, to the nearest minute, for 140 runners to complete a fun run.



Use the histogram to calculate the number of runners who took between 78.5 and 90.5 minutes to complete the fun run. **(5)**

?

Q2 The following table summarises the distances, to the nearest km, that 134 examiners travelled to attend a meeting in London.

Distance (km)	Number of examiners
41–45	4
46–50	19
51–60	53
61–70	37
71–90	15
91–150	6

(*a*) Give a reason to justify the use of a histogram to represent these data.

? **(1)**

(*b*) Calculate the frequency densities needed to draw a histogram for these data.

(DO NOT DRAW THE HISTOGRAM)

(2)

?

Q3 **[May 2013 (R) Q3]** An agriculturalist is studying the yields, y kg, from tomato plants. The data from a random sample of 70 tomato plants are summarised below.

Yield (y kg)	Frequency (f)	Yield midpoint (x kg)
$0 \leq y < 5$	16	2.5
$5 \leq y < 10$	24	7.5
$10 \leq y < 15$	14	12.5
$15 \leq y < 25$	12	20
$25 \leq y < 35$	4	30

(You may use $\sum fx = 755$ and $\sum fx^2 = 12\,037.5$)

A histogram has been drawn to represent these data.
The bar representing the yield $5 \leq y < 10$ has a width of 1.5 cm and a height of 8 cm.

(*a*) Calculate the width and the height of the bar representing the yield $15 \leq y < 25$. **(3)**
(*b*) Use linear interpolation to estimate the median yield of the tomato plants. **(2)**
(*c*) Estimate the mean and the standard deviation of the yields of the tomato plants. **(4)**
(*d*) Describe, giving a reason, the skewness of the data. **(2)**

(a) ? **(3)**

(b) ? **(2)**

(c) ? **(4)**

(d) ? **(2)**

Q4 [June 2007 Q5]

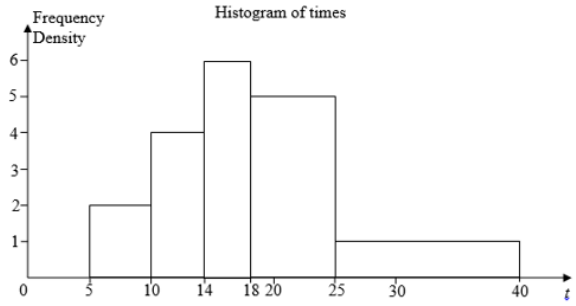


Figure 2 shows a histogram for the variable t which represents the time taken, in minutes, by a group of people to swim 500 m.

(*a*) Copy and complete the frequency table for t.

t	5 – 10	10 – 14	14 – 18	18 – 25	25 – 40
Frequency	10	16	24		

(2)

(*b*) Estimate the number of people who took longer than 20 minutes to swim 500 m. **(2)**

(*c*) Find an estimate of the mean time taken. **(4)**

(*d*) Find an estimate for the standard deviation of t. **(3)**

(*e*) Find the median and quartiles for t. **(4)**

One measure of skewness is found using $\frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$.

~~(*f*) Evaluate this measure and describe the skewness of these data.~~ ~~**(2)**~~

5(a) ?

(b) ?

(c) ?

(d) ?

(e) ?

(f) ?

Q5

[Jan 2013 Q5] A survey of 100 households gave the following results for weekly income £y.

Income y (£)	Mid-point	Frequency f	
$0 \leq y < 200$	100	12	
$200 \leq y < 240$	220	28	
$240 \leq y < 320$	280	22	
$320 \leq y < 400$	360	18	
$400 \leq y < 600$	500	12	
$600 \leq y < 800$	700	8	

(You may use $\sum fy^2 = 12\,452\,800$)

A histogram was drawn and the class $200 \leq y < 240$ was represented by a rectangle of width 2 cm and height 7 cm.

(*a*) Calculate the width and the height of the rectangle representing the class $320 \leq y < 400$ **(3)**

(*b*) Use linear interpolation to estimate the median weekly income to the nearest pound. **(2)**

(*c*) Estimate the mean and the standard deviation of the weekly income for these data. **(4)**

One measure of skewness is $\frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$.

~~(*d*) Use this measure to calculate the skewness for these data and describe its value.~~ ~~**(2)**~~

(a) ?

(b) ?

(c) ?

(d) ?

Q6 **[May 2010 Q5]** A teacher selects a random sample of 56 students and records, to the nearest hour, the time spent watching television in a particular week.

Hours	1–10	11–20	21–25	26–30	31–40	41–59
Frequency	6	15	11	13	8	3
Mid-point	5.5	15.5		28		50

(*a*) Find the mid-points of the 21–25 hour and 31–40 hour groups. **(2)**

A histogram was drawn to represent these data. The 11–20 group was represented by a bar of width 4 cm and height 6 cm.

(*b*) Find the width and height of the 26–30 group. **(3)**

(*c*) Estimate the mean and standard deviation of the time spent watching television by these students. **(5)**

(*d*) Use linear interpolation to estimate the median length of time spent watching television by these students. **(2)**

The teacher estimated the lower quartile and the upper quartile of the time spent watching television to be 15.8 and 29.3 respectively.

~~(*e*) State, giving a reason, the skewness of these data. **(2)**~~

(a) ?

(b) ?

(c) ?

(d) ?

(e) ?

Q7 **[Jan 2009 Q5]** In a shopping survey a random sample of 104 teenagers were asked how many hours, to the nearest hour, they spent shopping in the last month. The results are summarised in the table below.

Number of hours	Mid-point	Frequency
0 – 5	2.75	20
6 – 7	6.5	16
8 – 10	9	18
11 – 15	13	25
16 – 25	20.5	15
26 – 50	38	10

A histogram was drawn and the group (8 – 10) hours was represented by a rectangle that was 1.5 cm wide and 3 cm high.

(*a*) Calculate the width and height of the rectangle representing the group (16 – 25) hours. **(3)**

(*b*) Use linear interpolation to estimate the median and interquartile range. **(5)**

(*c*) Estimate the mean and standard deviation of the number of hours spent shopping. **(4)**

~~(*d*) State, giving a reason, the skewness of these data. **(2)**~~

~~(*e*) State, giving a reason, which average and measure of dispersion you would recommend to use to summarise these data. **(2)**~~

(a) a ?

(b) b ?

(c) c ?

(d) d ?

(e) e ?

(5)